2. PRD — TCC Event Clearance App

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A mobile-first installable web app (PWA) where students show a rotating QR, department SBOs scan it offline, and signed activity cards and attendance sheets are produced automatically for about 6,000 students in 8 departments.

Roles and permissions

Three roles use the app. Instructors have no account.

Capability	Student	Department SBO officer	Head officer
Register with Google, take live photo	Yes	Yes	Yes
Show rotating QR, view and export own card	Yes	No	No
Approve registrations and device resets	No	Own department	All departments
Scan QR, reject photo, manual check-in	No	Yes, at any venue	Yes
Check card (signed or missing per slot)	No	Yes	Yes
Draft events and card templates	No	Yes	Yes
Publish events (after instructor approval)	No	No	Yes
Draw and manage own signature	No	Yes	Yes
Add or remove officers, assign positions	No	No	Yes
View, edit, export and print attendance	Own record only	Own department	All departments

Features by role

Student

- Install gate, Google sign-in, profile (name, school ID, department, year level, section) and consent screen.
- Live camera photo with an on-device face check.
- Event list with a one-time download of each event pack (card design, details, venues).
- Rotating QR that works offline, with an optional location read when the event requires it.
- Digital activity card showing a signed stamp for every slot; export as image or PDF.
- Device-reset request when changing phones.

SBO officer

- Signature pad: draw once, saved to the account, redrawable any time.
- Event builder: name, days, slots per day, venues, eligible departments, signer name and position per slot, card template, "Approved by" and date.
- AI artwork generation for the card background, with a regeneration limit per event and preset templates as a backup.
- Scanner: pick venue and slot, scan the student's QR, see photo, name, department and section, then accept or reject with a reason.
- Manual check-in by school ID or name, logged as manual.
- Check card: scan the card's QR or search a student to see every slot as signed or missing.
- Approve registrations and device resets for the officer's own department.

Head officer

- Everything an officer can do, across all departments.
- Publish events, manage officers and positions, revoke a lost scanner phone.

Web dashboard (officers only)

- Attendance sorted by department, year level and section, filterable by event, day, slot, venue and officer.
- Edit names or records (every edit is logged), void a duplicate, resolve a dispute.
- Export to Excel and PDF, print attendance sheets and activity cards.

Anti-cheating requirements

The goal is to make cheating harder than attending. No web app can block every case.

ID	Requirement	Release
AC1	Live camera photo only (no gallery); the device checks for exactly one clear, centered face	MVP
AC2	The scanner shows the photo large, with name, department and section, so the SBO can compare it to the person	MVP
AC3	The SBO can reject with a reason; a rejection forces a retake and blocks the QR until the new photo is taken	MVP
AC4	The photo is locked after acceptance; changes go through an SBO	MVP
AC5	One account, one device; a new phone needs a reset approved by the department SBO	MVP
AC6	The QR changes every 30 seconds and is signed by the device key	MVP
AC7	One valid check-in per student per slot; replays and double scans are flagged	MVP
AC8	Optional location check, off by default; flag-only or block when on	MVP (optional)
AC9	Install gate: the app is unusable until installed	MVP
AC10	Every stamp is signed by the scanner key; cards are verified from records on the SBO Check card screen, never by appearance	MVP
AC11	Audit log of every scan and admin action	MVP
AC12	Dashboard flags unusual patterns (many scans from one device, far-from-venue locations)	Phase 2
AC13	School policy: lending an account or QR is a violation with a consequence	Policy

Honest limit: two people who share a live screen can still defeat the QR and location checks. The photo comparison at the scanner is what catches them.

Offline and scale requirements

Many students have no mobile data, so everything on event day must work without internet. Targets below are proposals to test, not measurements.

Requirement	Target
Scan to accepted	2 seconds or less per student
Throughput per scanner	600 to 1,000 students per hour
Scanners at intramurals	One per department, two for the largest (CCJPS, estimated 1,000 to 1,500 students)
Roster pack per scanner	10 to 25 MB, filtered to that department
Works offline	QR generation, scanning, manual check-in, card display, Check card from cached records, signature pad
Needs internet	Install and registration, downloading event packs and rosters, creating events and AI artwork, uploading scans
Sync	Scans upload when signal returns; the scanner shows "X scans waiting to sync" and a green "ready for offline" state
Safety net	Export unsynced scans as a file if sync fails
Clock tolerance	Accept small phone clock differences and show "this phone's clock is off" instead of a generic error
Stamp delivery	By sync by default; an optional receipt-QR handshake gives the student the stamp immediately
Fallback	Manual check-in by school ID for dead phones, broken cameras and students without smartphones

MVP and Phase 2

Area	MVP	Phase 2
Accounts	Google sign-in, registration, department approval, device binding, reset requests	Bulk import from the registrar if the school agrees
Events	Multi-day, multi-slot, multi-venue events with per-slot signers	Reusable event copies and analytics
Card	Template-driven card, signature stamps, image and PDF export	AI artwork variations, more preset templates
Check-in	Rotating QR, offline scanner, manual check-in, optional location flag	Receipt-QR handshake tuning, anomaly flags
Checking	SBO Check card screen	Dispute workflow polish
Web dashboard	Attendance by department, year and section; Excel and PDF export; print	Editing history view, charts
Notifications	None	Push reminders for installed apps
Platforms	PWA on Android and iPhone	Store-wrapped app if the school funds it

Success metrics and out of scope

Proposed metrics, to confirm after the pilot

- Average time from scan to accepted: 2 seconds or less.
- Peak waiting time at any scanner: 3 minutes or less.
- Attendance records uploaded within 24 hours of the event: 100 percent, with none lost.
- Students registered and approved before intramurals: 90 percent or more.
- Students needing manual check-in: under 5 percent.

Out of scope for version 1

- Registrar integration, instructor accounts, face recognition, native store apps, payments and grading.